# Lista de Exercícios 2.1

INFERÊNCIA ESTATÍSTICA NÃO PARAMÉTRICA - 2026.1

Milena Paz Freitas
2026-03-25

## Enunciado

Gere amostras das distribuições abaixo. Para cada amostra, construa 3 gráficos: o histograma de densidade mais a função densidade de probabilidade (caso contínuo), função distribuição (empírica e teórica) e função quantil (empírica e teórica). Para as variáveis discretas, fazer apenas gráfico de barras das frequências relativas ou histograma de densidade (mais a função de probabilidade) e função distribuição (empírica e teórica).

a. X ~ N(5, 2), n=50.

b. X ~ N(5, 2), n=200.

c. X ~ N(100, 1), n=50.

d. X ~ exp(5), n=50.

e. X ~ exp(1/5), n=50.

f. X ~ exp(5), n=200.

g. X ~ Poisson(5), n=50.

h. X ~ Poisson(5), n=100.

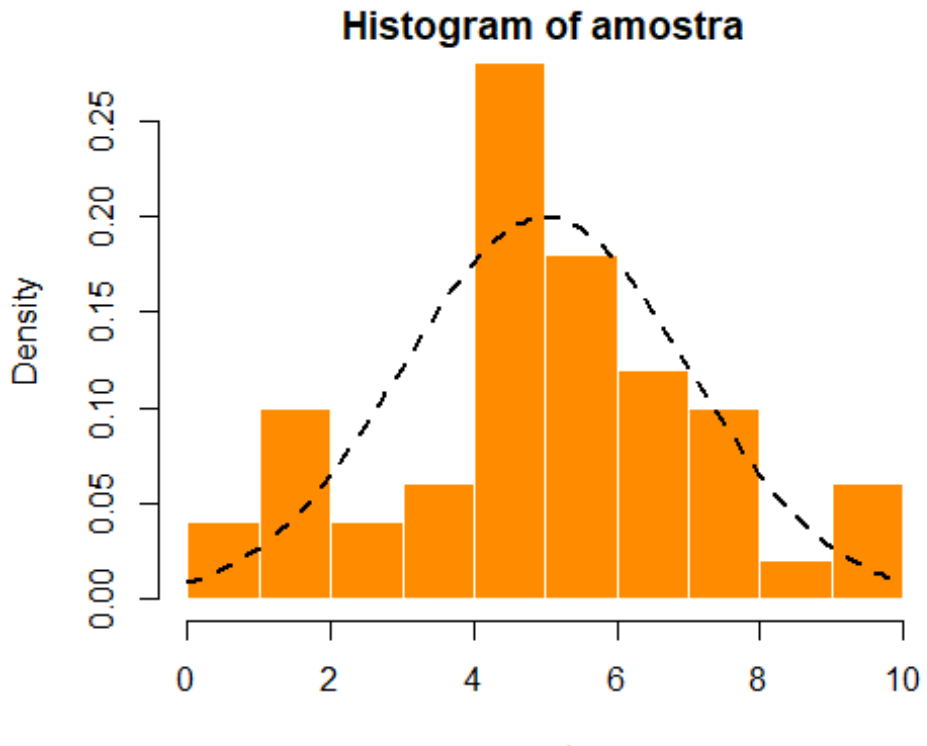
## Resolução

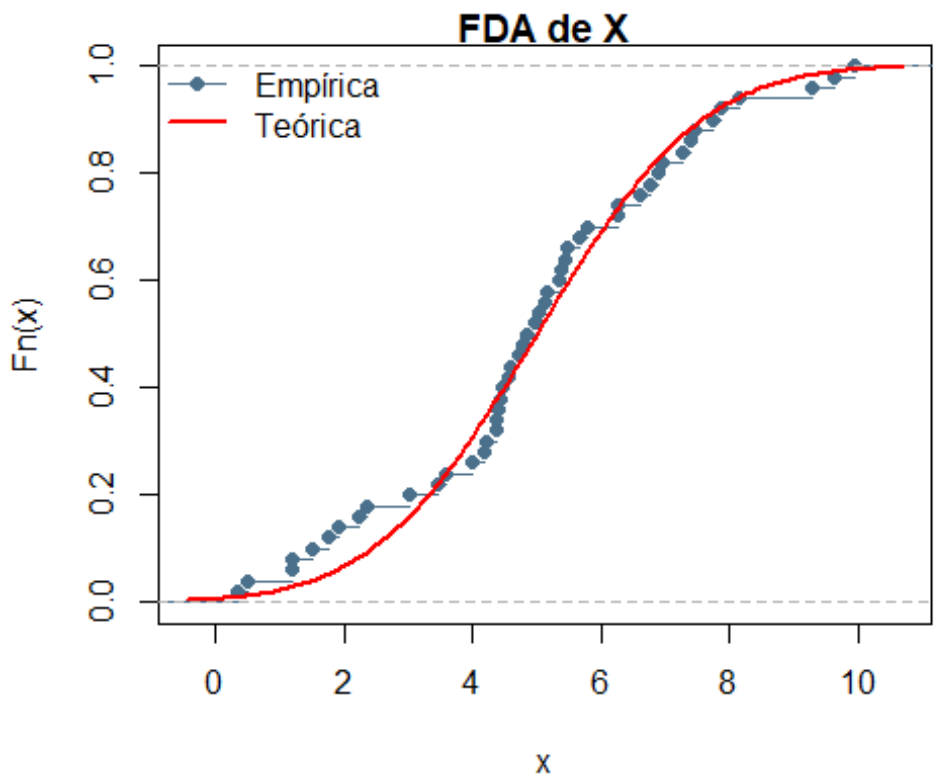
```r
# cria-se uma função que realize a geração de amostra e gráficos
sim <- function(n=50,distrib,theta){
  par(mar=c(4.1,4.1,1,1))
   if(distrib=="norm"){
    amostra <- rnorm(n,theta[1],theta[2])
    hist(amostra,freq=F,col="darkorange",border="white")
    curve(dnorm(x,theta[1],theta[2]),add=T,lwd=2,lty=2)
    plot(ecdf(amostra),col="skyblue4",main="FDA de X")
    curve(pnorm(x,theta[1],theta[2]),add=T,col='red',lwd=2)
    legend("topleft",legend=c("Empírica","Teórica"),
           col=c("skyblue4","red"),pch=c(19,NA),lwd=c(1,2),bty="n")
```

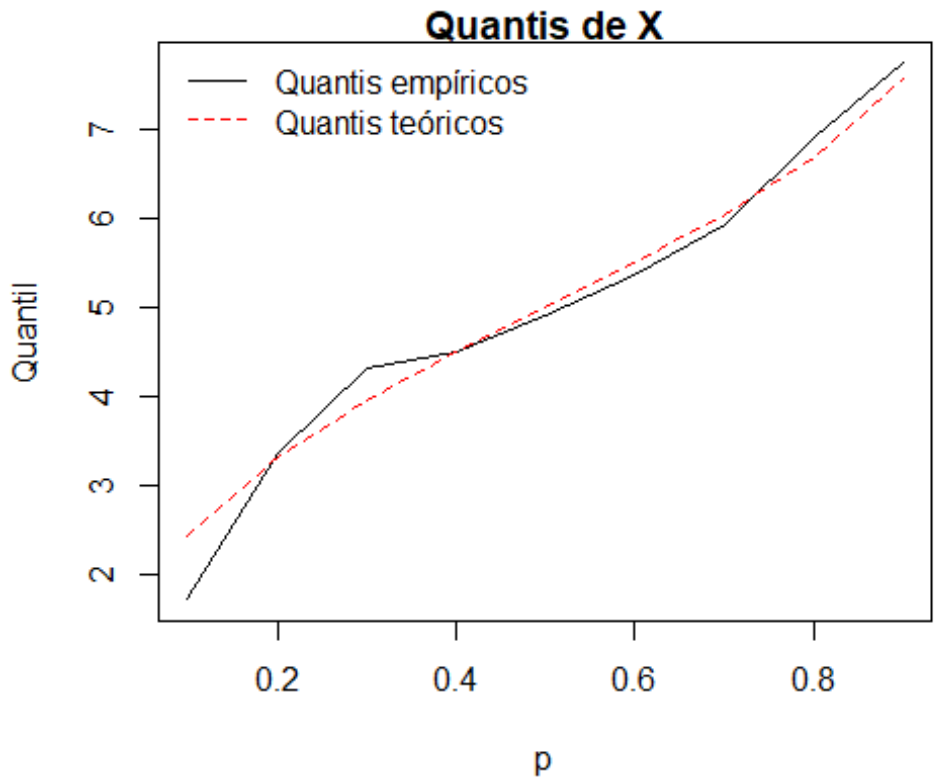
```
    plot(seq(0.1,.9,0.1),quantile(amostra,probs=seq(0.1,.9,0.1)),
           type="l", main="Quantis de X",xlab="p",ylab="Quantil")
    lines(seq(0.1,.9,0.1),qnorm(seq(0.1,0.9,0.1), theta[1],theta[2]),lty=
2,col="red")
    legend("topleft",legend=c("Quantis empíricos","Quantis teóricos"),
             col=c("black","red"),lty=c(1,2),bty="n")
  }
  else{
    amostra <- do.call(paste0("r",distrib),args=list(n,theta))
    if(distrib=="exp"){
        hist(amostra,freq=F,col="darkorange",border="white",breaks=15)
        curve(dexp(x,theta),add=T,lwd=2,lty=2)
        plot(ecdf(amostra),col="skyblue4",main="FDA de X")
        curve(pexp(x,theta),add=T,col='red',lwd=2)
        legend("topleft",legend=c("Empírica","Teórica"),
             col=c("skyblue4","red"),pch=c(19,NA),lwd=c(1,2),bty="n")
        plot(seq(0.1,.9,0.1),quantile(amostra,probs=seq(0.1,.9,0.1)),
             type="l", main="Quantis de X",xlab="p",ylab="Quantil")
        lines(seq(0.1,.9,0.1),qexp(seq(0.1,0.9,0.1), theta),lty=2,col="re
d")
        legend("topleft",legend=c("Quantis empíricos","Quantis teóricos")
,col=c("black","red"),lty=c(1,2),bty="n")
      }
      else{
        hist(amostra,freq=F,col="darkorange",border="white",breaks=15)
        lines(seq(0,max(amostra),1),dpois(seq(0,max(amostra),1),theta),lw
d=2,lty=2)
        plot(ecdf(amostra),col="skyblue4",main="FDA de X")
        lines(seq(0,max(amostra),1),ppois(seq(0,max(amostra),1),theta),co
l='red',lwd=2,type="s")
        legend("topleft",legend=c("Empírica","Teórica"),
             col=c("skyblue4","red"),pch=c(19,NA),lwd=c(1,2),bty="n")
      }
    }
}
```

## A.

```
sim(distrib="norm", theta=c(5,2))
```

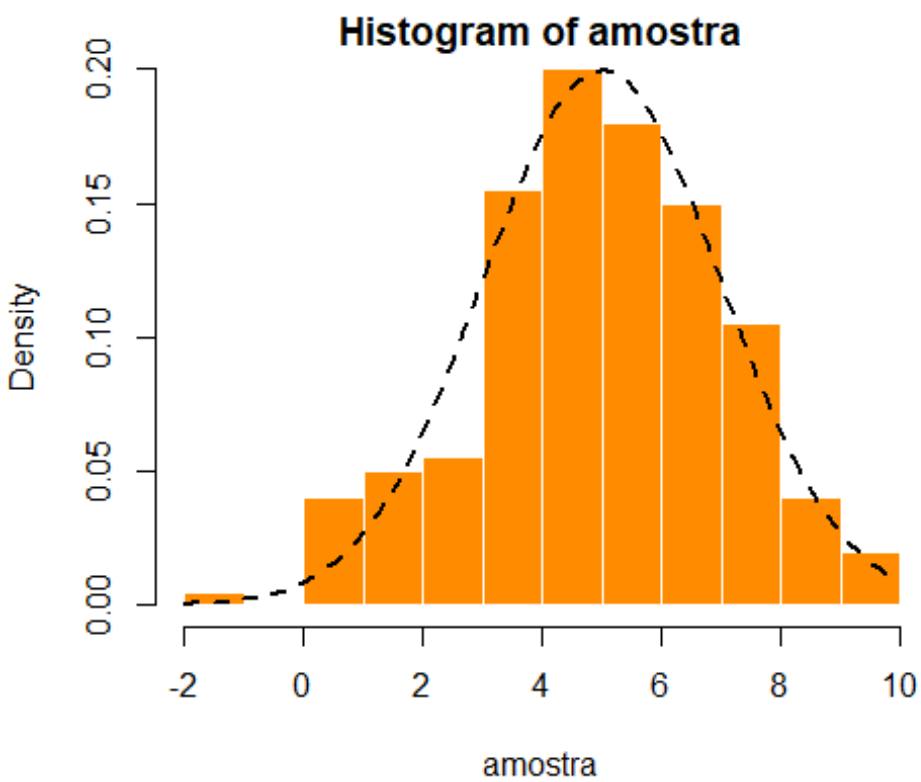

Histogram of amostra
Density
0.00
0.05
0.10
0.15
0.20
0.25
0
2
4
6
8
10
amostra

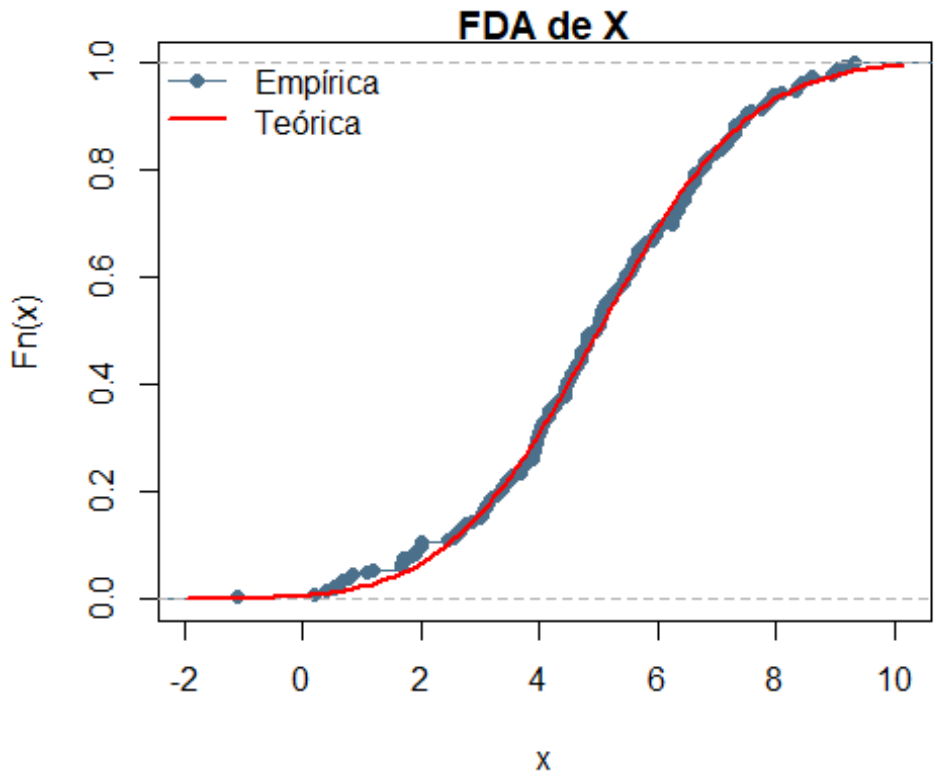

FDA de X
Empírica
Teórica
Fn(x)
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
x

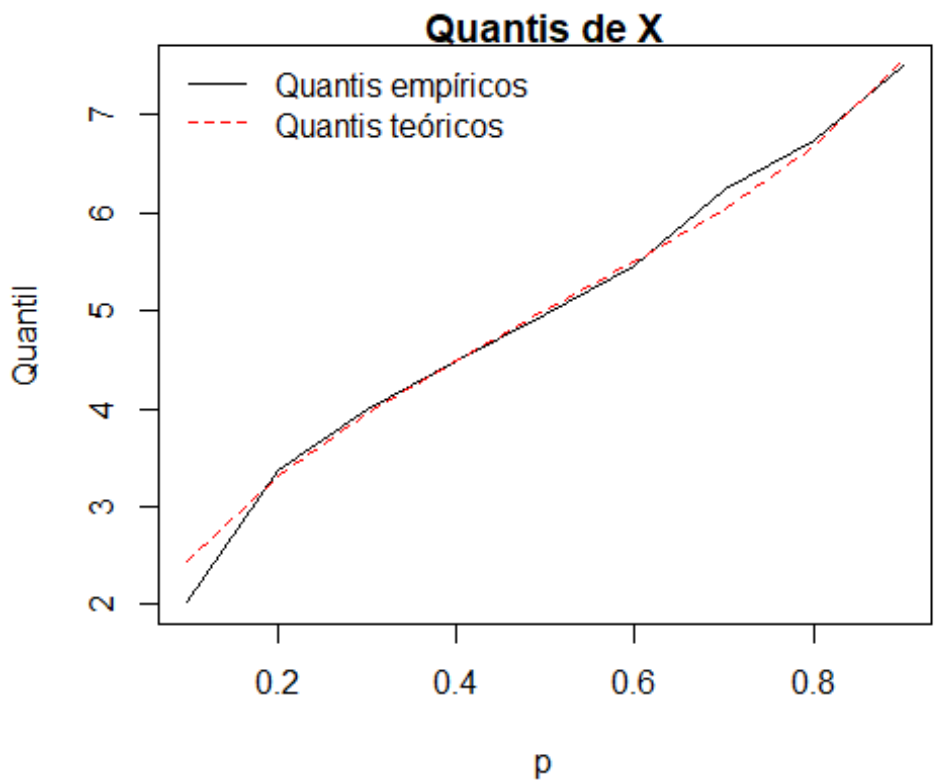



B.

```
sim(n=200,distrib="norm", theta=c(5,2))
```

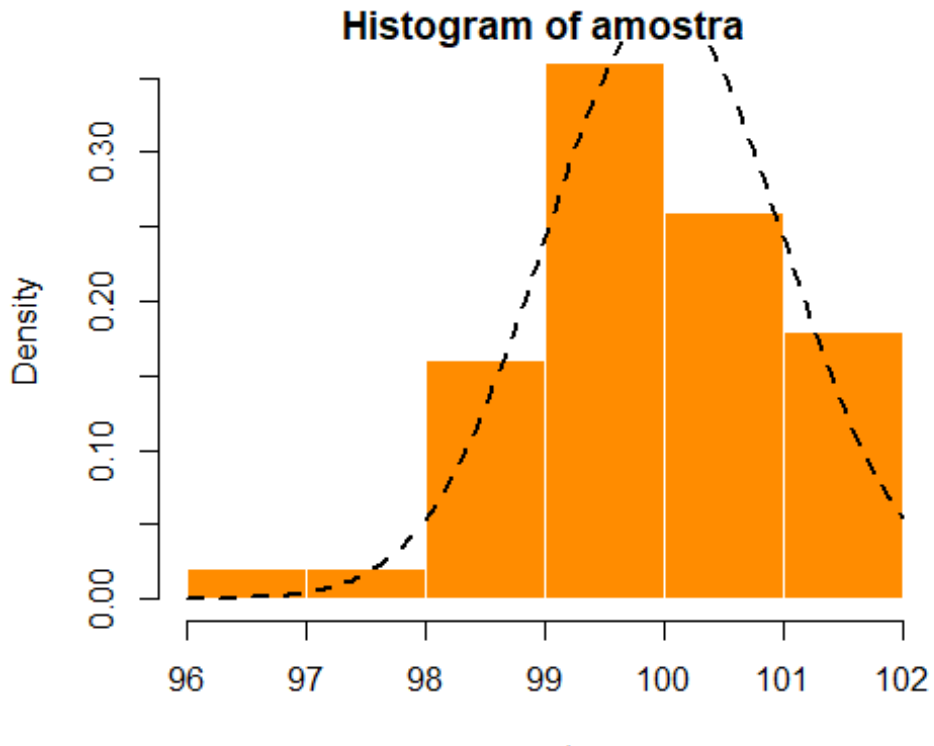


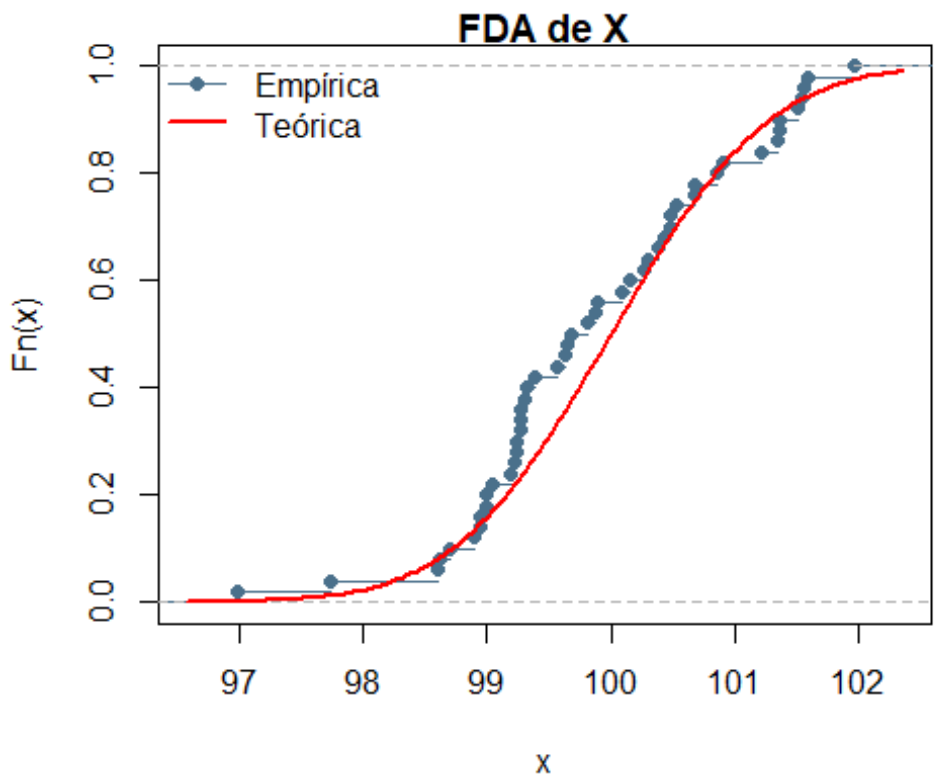


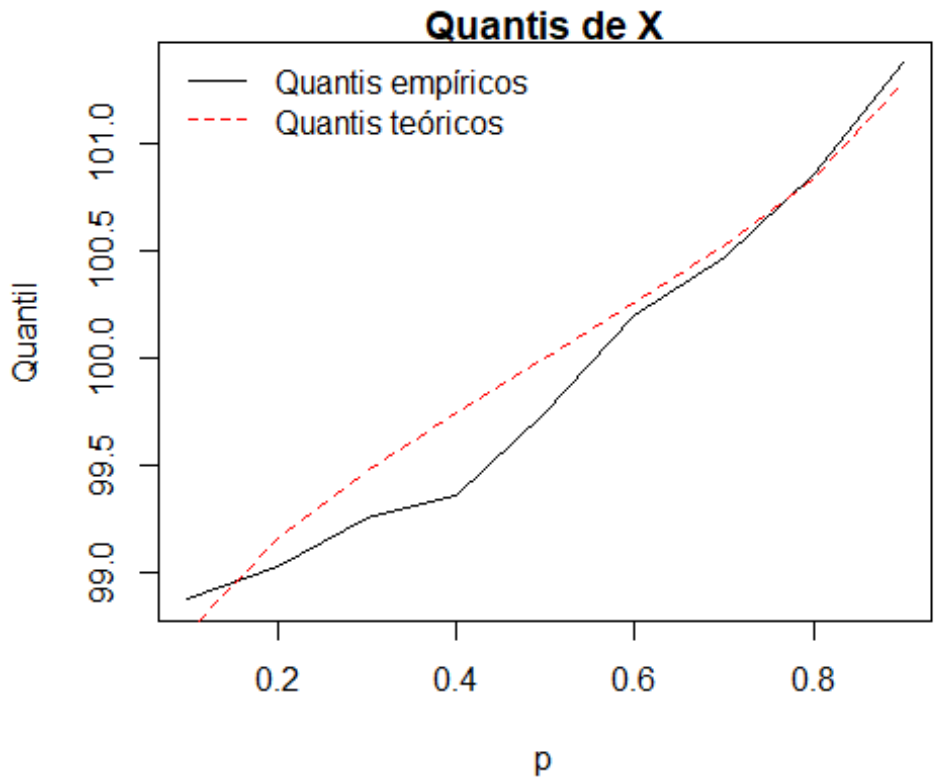




## C.

```
sim(distrib="norm", theta=c(100,1))
```

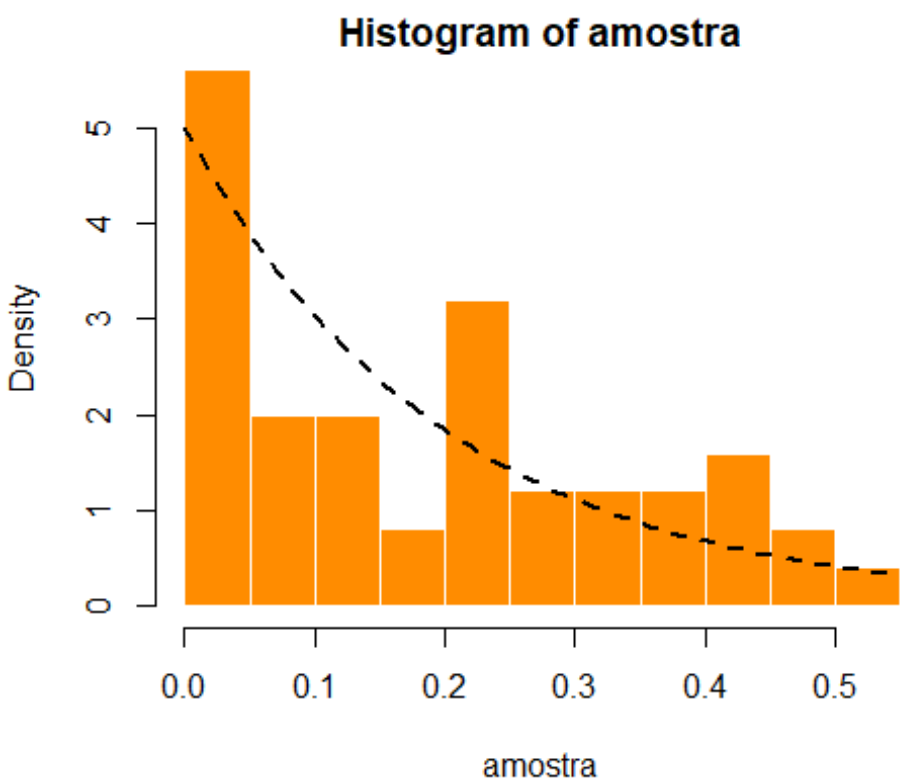

Histogram of amostra
Density
0.00
0.10
0.20
0.30
96
97
98
99
100
101
102
amostra

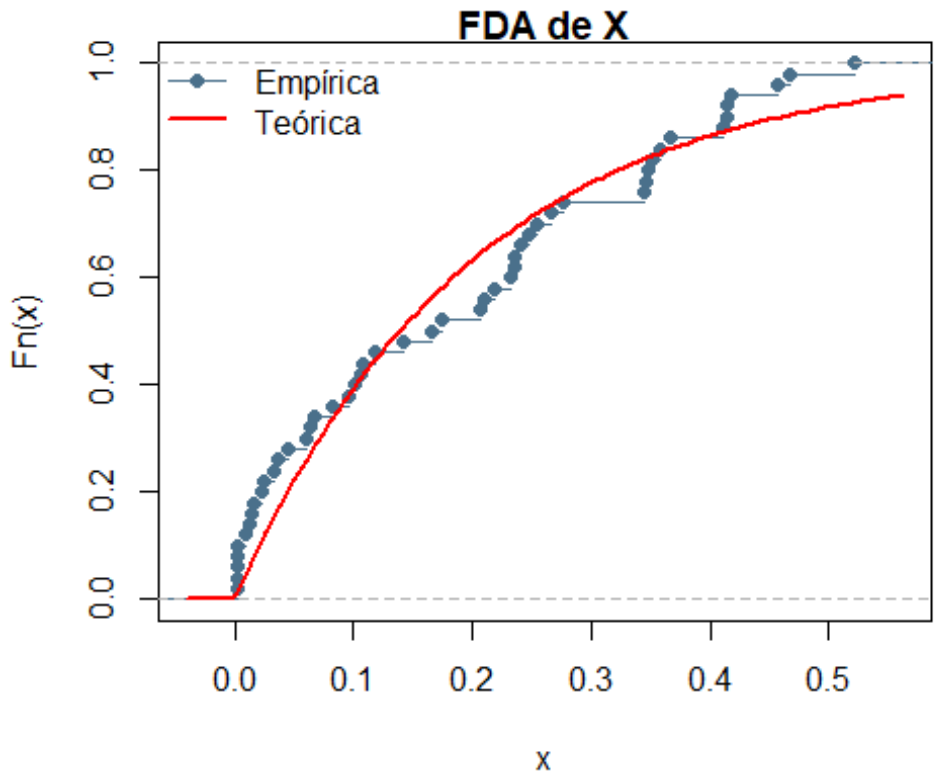

FDA de X
Empírica
Teórica
Fn(x)
0.0
0.2
0.4
0.6
0.8
1.0
97
98
99
100
101
102
x

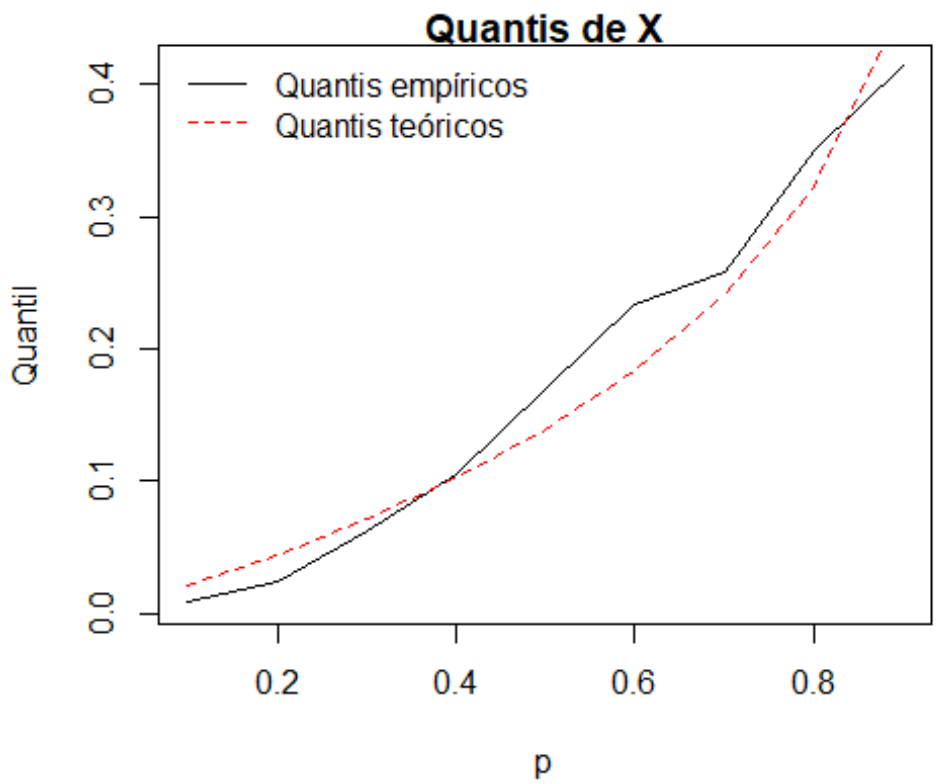



## D.

```
sim(distrib="exp", theta=5)
```

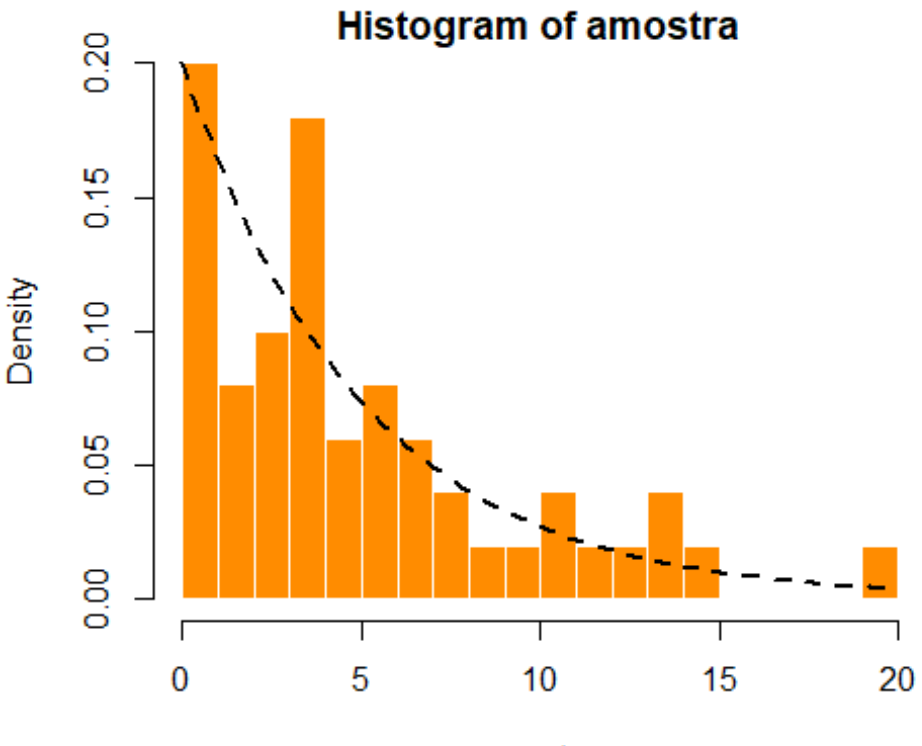


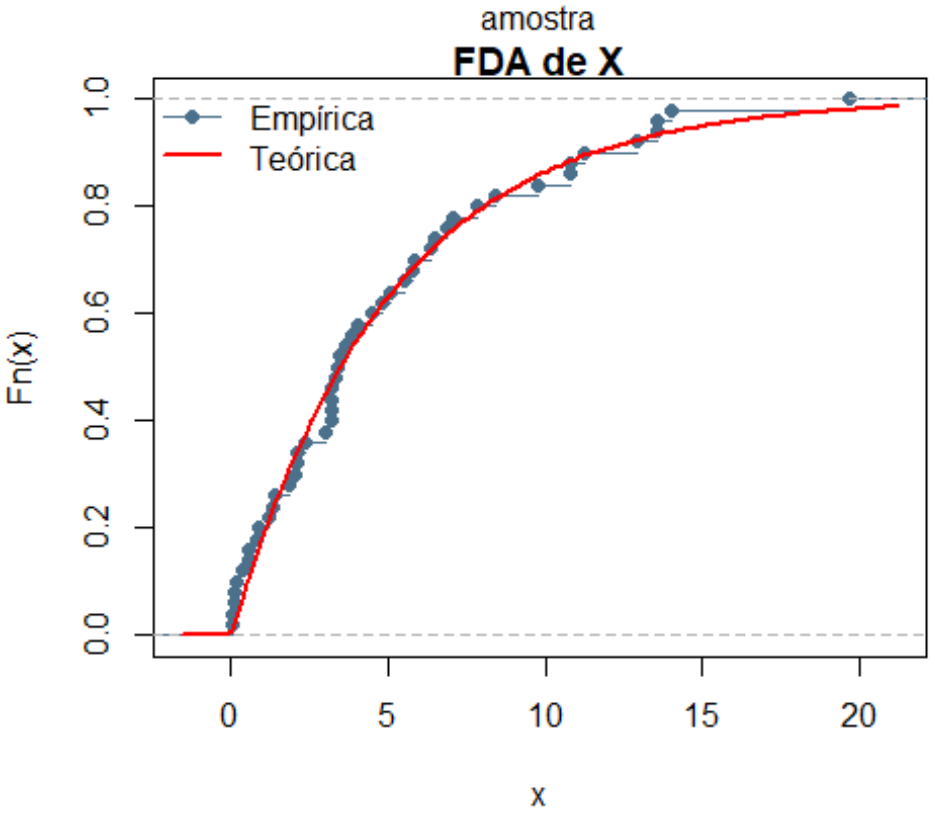


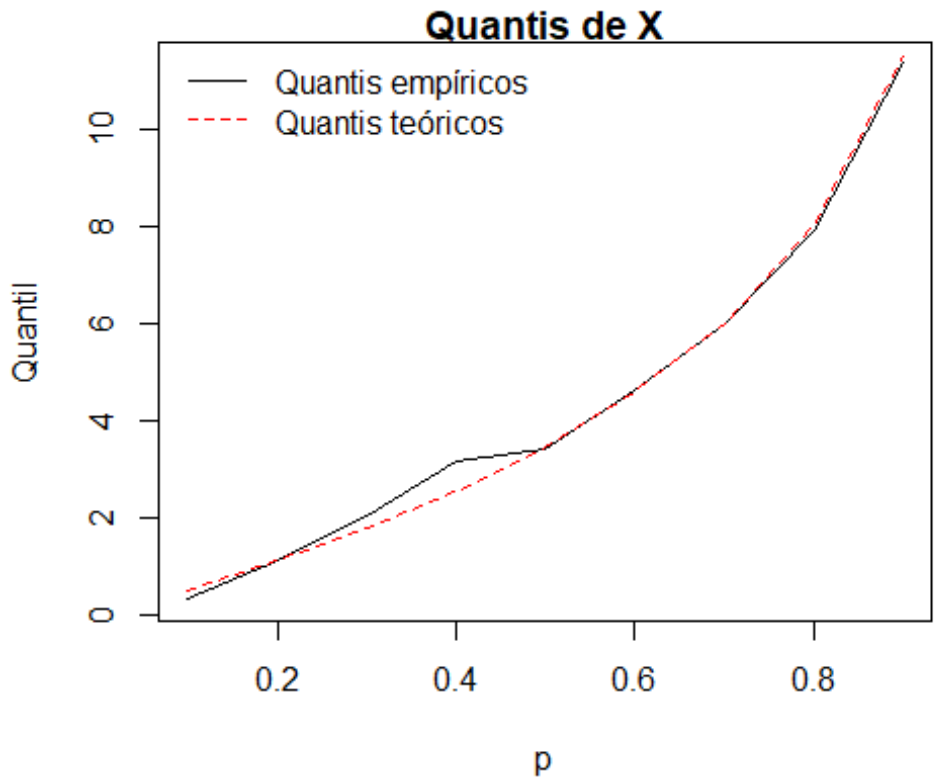




## E.

```
sim(distrib="exp", theta=1/5)
```

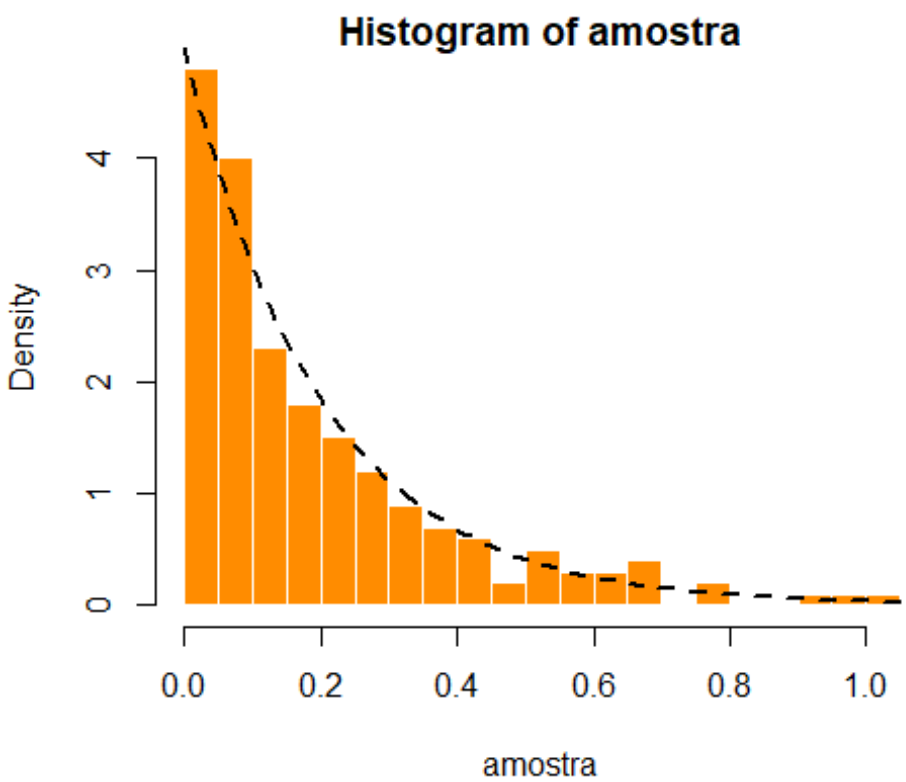

Histogram of amostra
Density
0.00
0.05
0.10
0.15
0.20
0
5
10
15
20
amostra

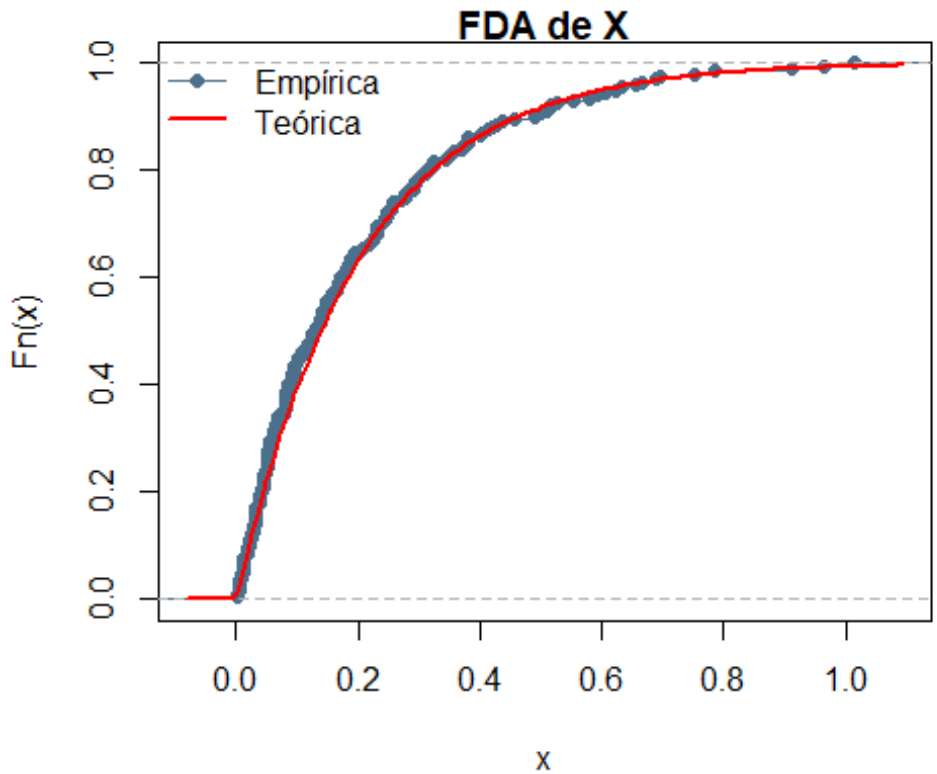

FDA de X
Empírica
Teórica
Fn(x)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
x

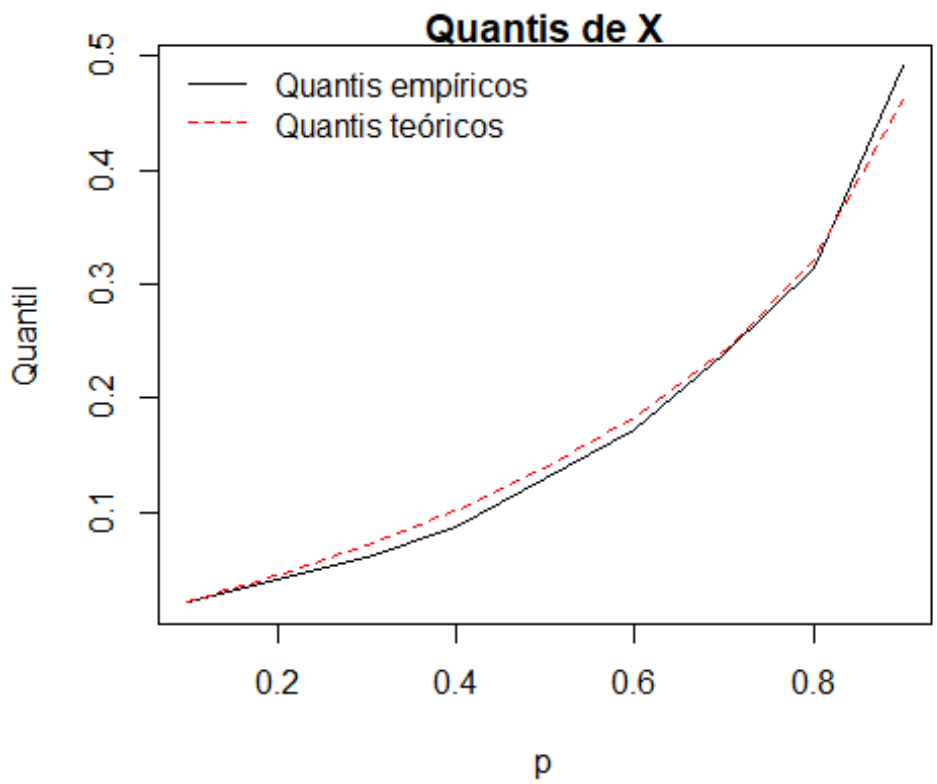



F.

```
sim(n=200,distrib="exp", theta=5)
```

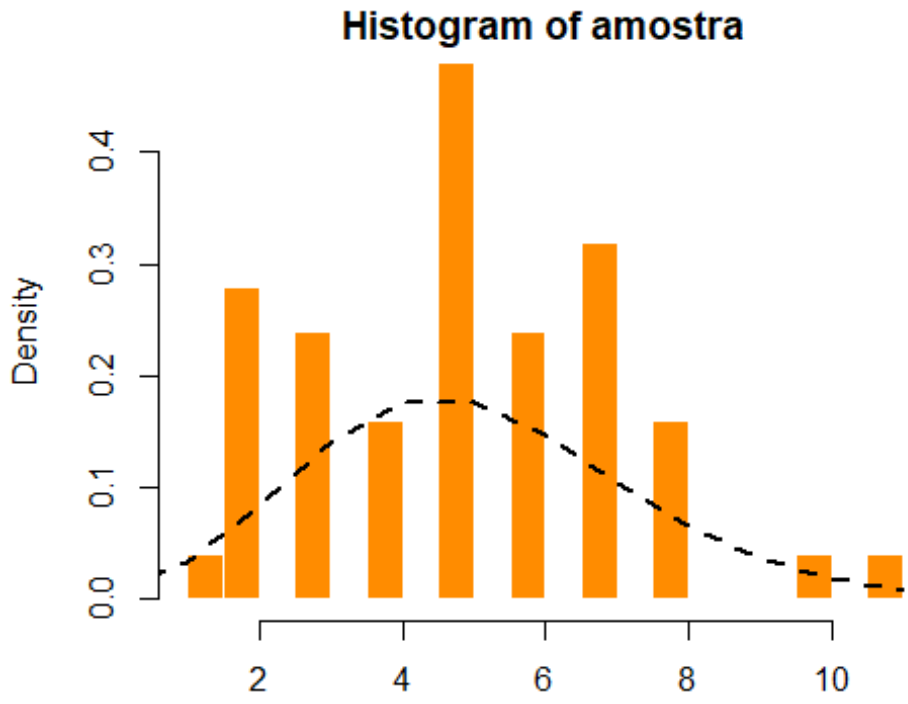


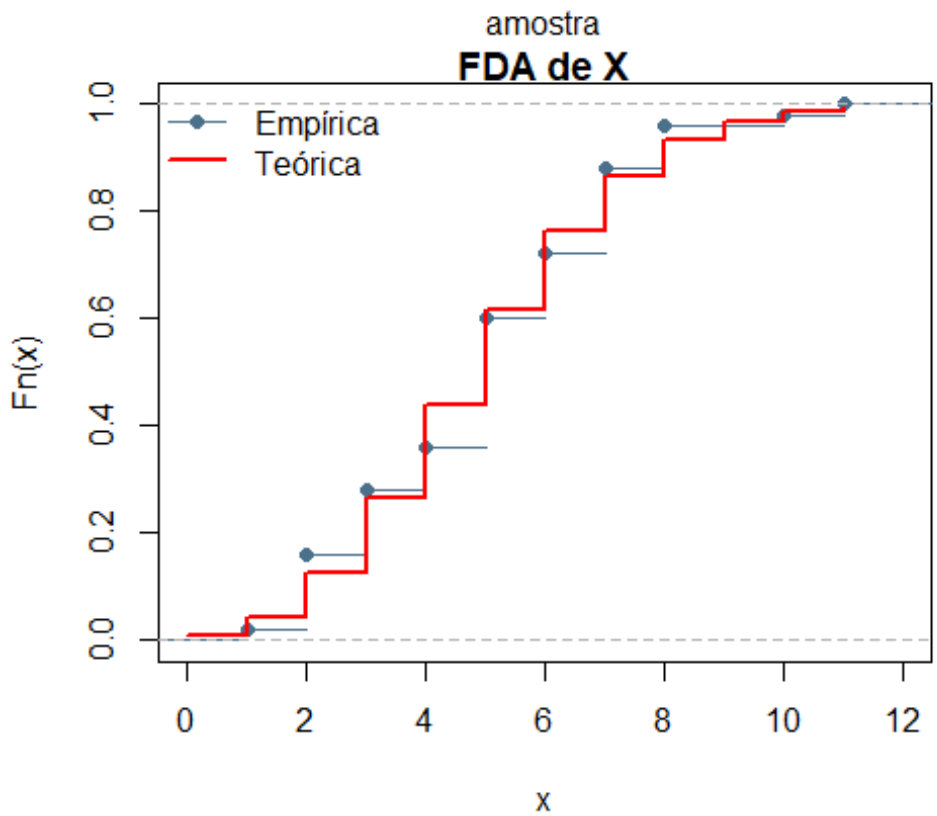



## G.

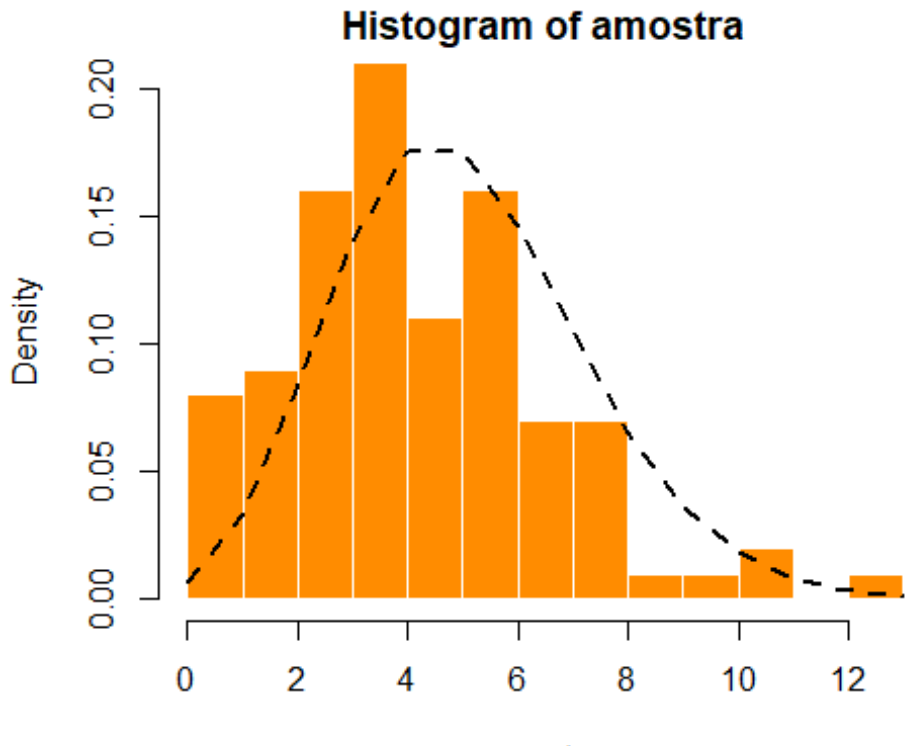
```
sim(distrib="pois", theta=5)
```

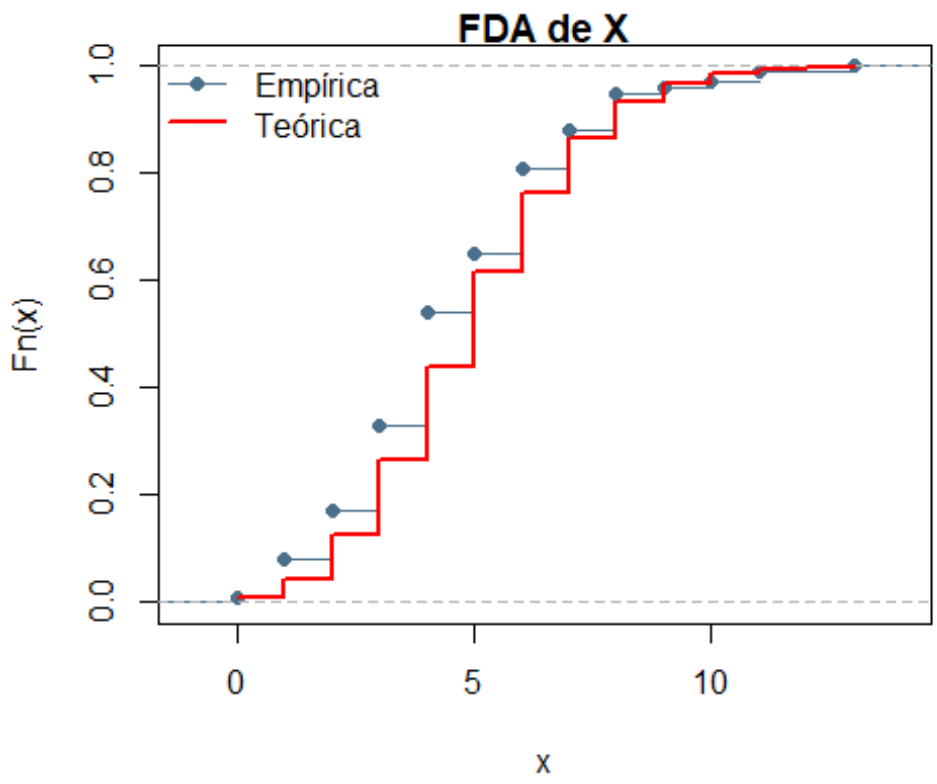



## H.

```
sim(n=100,distrib="pois", theta=5)
```


Histogram of amostra
Density
0.00
0.05
0.10
0.15
0.20
0
2
4
6
8
10
12
amostra


FDA de X
Empírica
Teórica
Fn(x)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
x